

Table 1: Workforce Characteristics, Janitors, 2001-2025

Contractor = Outsourced; In-House = Directly Employed

	2001-2005		2006-2010		2011-2015		2016-2019		2021-2025	
	In-house	Contractor	In-house	Contractor	In-house	Contractor	In-house	Contractor	In-house	Contractor
Employment and wages										
Mean hourly wage (2025\$)	\$18.58	\$16.04	\$18.80	\$16.51	\$18.69	\$16.26	\$19.32	\$17.70	\$20.26	\$18.49
Unionization rate	20.8%	10.7%	20.5%	9.6%	18.4%	8.9%	16.9%	8.4%	16.4%	9.7%
Part-time rate	20.1%	27.9%	18.9%	28.6%	20.3%	29.2%	19.7%	26.2%	19.3%	23.3%
Government employee	32.7%	1.0%	33.8%	1.8%	30.2%	1.8%	28.5%	1.5%	31.0%	1.5%
Demographics										
Mean age	43.5	39.2	44.8	40.5	45.6	42.2	45.8	42.6	46.0	42.6
Female	30.1%	47.8%	27.9%	46.8%	26.7%	47.2%	28.6%	48.8%	30.1%	52.5%
Black	18.3%	16.8%	18.2%	14.8%	16.9%	17.1%	17.8%	16.7%	16.9%	14.9%
Hispanic	22.8%	44.0%	25.3%	45.2%	25.7%	49.8%	27.6%	47.3%	28.1%	50.8%
Education										
Less than high school	31.9%	43.9%	27.1%	38.8%	23.0%	36.6%	21.5%	32.7%	18.4%	29.0%
High school / GED	46.4%	40.6%	48.2%	41.2%	47.7%	42.7%	48.6%	44.3%	48.5%	44.6%
Some college	18.0%	12.4%	20.4%	14.6%	23.8%	16.1%	23.4%	17.3%	24.8%	18.7%
Bachelor's or higher	3.7%	3.1%	4.4%	5.3%	5.5%	4.6%	6.5%	5.8%	8.3%	7.8%
Sample										
Unweighted N (regression sample)	11,796	2,669	10,850	2,640	10,533	2,889	7,752	2,243	6,752	2,232
Workers represented (thousands)	1,625	403	1,601	410	1,639	484	1,644	544	1,489	567

Data: CPS-ORG via IPUMS. Years: 2001-2025, 2020 excluded. Part-time = usual hours under 30. Weights: EARNWT. Statistics use the author's regression sample (non-missing log_wage and every model covariate); N = observations with non-missing wages and all model covariates. Workers represented is average monthly employment estimated from EARNWT on all sampled base-sample workers, including those excluded for missing wages or covariates: sum(EARNWT) divided by the number of distinct survey months (ym) for that occupation in the period, in thousands, rounded to the nearest thousand. FE: state x year and clustering by survey month (ym) apply to regression tables only; Tables 1-2 are descriptive. Stars: * p<.10, ** p<.05, *** p<.01, two-tailed.

Table 2: Workforce Characteristics, Guards, 2001-2025

Contractor = Outsourced; In-House = Directly Employed

	2001-2005		2006-2010		2011-2015		2016-2019		2021-2025	
	In-house	Contractor	In-house	Contractor	In-house	Contractor	In-house	Contractor	In-house	Contractor
Employment and wages										
Mean hourly wage (2025\$)	\$22.36	\$19.24	\$22.83	\$19.06	\$23.16	\$18.42	\$23.75	\$19.81	\$23.61	\$21.05
Unionization rate	17.5%	8.3%	15.3%	9.1%	17.7%	8.5%	19.4%	10.9%	17.8%	8.6%
Part-time rate	14.0%	10.9%	13.3%	11.0%	13.9%	12.7%	12.8%	10.8%	12.7%	13.5%
Government employee	32.8%	3.3%	32.7%	2.9%	34.4%	2.2%	35.1%	2.4%	32.4%	2.6%
Demographics										
Mean age	41.4	41.3	42.0	41.3	42.4	41.3	42.4	40.8	43.2	41.0
Female	24.0%	23.0%	22.9%	22.5%	21.8%	20.7%	22.7%	23.3%	26.2%	25.7%
Black	24.3%	32.7%	23.5%	33.7%	22.7%	31.9%	24.6%	34.8%	25.5%	40.1%
Hispanic	13.4%	12.1%	13.2%	15.0%	16.9%	17.4%	17.6%	17.3%	19.2%	18.7%
Education										
Less than high school	9.3%	12.6%	5.0%	8.9%	4.7%	7.0%	3.4%	5.8%	3.7%	5.1%
High school / GED	38.6%	44.9%	35.0%	45.0%	33.2%	39.9%	31.0%	38.0%	35.6%	45.0%
Some college	36.0%	32.3%	41.1%	36.0%	40.7%	39.7%	42.1%	42.4%	38.6%	34.9%
Bachelor's or higher	16.1%	10.3%	18.8%	10.0%	21.3%	13.4%	23.5%	13.8%	22.1%	15.0%
Sample										
Unweighted N (regression sample)	2,911	2,515	2,964	2,569	2,937	2,479	2,236	1,837	1,931	1,767
Workers represented (thousands)	417	387	467	429	483	440	509	459	460	476

Data: CPS-ORG via IPUMS. Years: 2001-2025, 2020 excluded. Part-time = usual hours under 30. Weights: EARNWT. Statistics use the author's regression sample (non-missing log_wage and every model covariate); N = observations with non-missing wages and all model covariates. Workers represented is average monthly employment estimated from EARNWT on all sampled base-sample workers, including those excluded for missing wages or covariates: sum(EARNWT) divided by the number of distinct survey months (ym) for that occupation in the period, in thousands, rounded to the nearest thousand. FE: state x year and clustering by survey month (ym) apply to regression tables only; Tables 1-2 are descriptive. Stars: * p<.10, ** p<.05, *** p<.01, two-tailed.

Table 3: Author's Extension of the Outsourcing Wage Penalty by Period, 2001-2025
Outsourced Coefficient on Log Wage, Janitors and Guards; SE in Parentheses

	Janitors			Guards		
	Coef		N	Coef		N
2001-2003	-0.042 (0.013)	***	8,884	-0.095 (0.013)	***	3,316
2004-2006	-0.056 (0.012)	***	8,330	-0.104 (0.013)	***	3,170
2007-2009	-0.039 (0.012)	***	8,035	-0.094 (0.015)	***	3,262
2010-2012	-0.041 (0.011)	***	8,093	-0.118 (0.016)	***	3,518
2013-2015	-0.040 (0.011)	***	8,035	-0.147 (0.018)	***	3,108
2016-2019	-0.017 (0.008)	**	9,995	-0.091 (0.015)	***	4,073
2021-2025	-0.028 (0.011)	**	8,984	-0.057 (0.015)	***	3,696
Pooled 2001-2025	-0.035 (0.004)	***	60,356	-0.097 (0.006)	***	24,143

Data: CPS-ORG via IPUMS. Author's Extension: the author's wage regression estimated separately for each period, 2001-2025 (2020 excluded). Weights: EARNWT. FE: state x year. Clustering: survey month (ym). Stars: * p<.10, ** p<.05, *** p<.01, two-tailed. No reported comparison values exist for these periods.

Table 1: Table 4: Union Mediation of the Outsourcing Wage Penalty, 2001-2025

Period	Union Rate In-House	Union Rate Out- sourced	Gap	Union Premium (β)	Raw Penalty	Union- Gap Compo- nent (log points)	% Explained	N
Janitors								
2001-2005	20.8%	10.7%	10.1pp	0.167 (0.010)	-0.061 (0.009)	-0.017 (0.002)	27.6%	14,46
2006-2010	20.5%	9.6%	10.9pp	0.151 (0.010)	-0.054 (0.009)	-0.016 (0.002)	30.5%	13,49
2011-2015	18.4%	8.9%	9.5pp	0.159 (0.012)	-0.054 (0.009)	-0.015 (0.002)	28.1%	13,42
2016-2019 [†]	16.9%	8.4%	8.5pp	0.117 (0.012)	-0.026 (0.008)	-0.010 (0.001)	38.0%	9,99
2021-2025	16.4%	9.7%	6.7pp	0.089 (0.013)	-0.033 (0.011)	-0.006 (0.001)	18.2%	8,98
<i>Change, 2001-2005 to 2021-2025</i>					+0.028 (0.014)	+0.011 (0.002)		
Guards								
2001-2005	17.5%	8.3%	9.3pp	0.144 (0.017)	-0.108 (0.010)	-0.013 (0.002)	12.4%	5,42
2006-2010	15.3%	9.1%	6.2pp	0.155 (0.019)	-0.113 (0.013)	-0.010 (0.002)	8.4%	5,53
2011-2015	17.7%	8.5%	9.3pp	0.160 (0.020)	-0.151 (0.013)	-0.015 (0.003)	9.8%	5,41
2016-2019 [†]	19.4%	10.9%	8.5pp	0.144 (0.021)	-0.104 (0.015)	-0.012 (0.003)	11.7%	4,07
2021-2025	17.8%	8.6%	9.2pp	0.050 (0.026)	-0.061 (0.015)	-0.005 (0.003)	7.6%	3,69
<i>Change, 2001-2005 to 2021-2025</i>					+0.047 (0.018)	+0.009 (0.003)		

Gap = union rate, in-house minus outsourced, in percentage points. Union premium (β) is the union coefficient from the author's base model. Raw penalty is the outsourced coefficient from the nonunion model (no union control). Union-gap component (log points) = $-(\text{gap} \times \text{union premium})$, with an approximate delta-method SE: $\sqrt{\beta^2 \text{var}(\text{gap}) + \text{gap}^2 \text{var}(\beta)}$; shown as negative, in the same direction as the raw penalty. % explained = $|\text{union-gap component}| / |\text{raw penalty}|$, and is unstable where the raw penalty is small. The change row compares 2001-2005 to 2021-2025 (independent bins, $\text{SE} = \sqrt{\text{SE}_1^2 + \text{SE}_2^2}$). SEs in parentheses. Descriptive decomposition, not causal. Data: CPS-ORG via IPUMS.

Table 5: Minimum Wage Interaction, Leads, and Trend, 2001-2025
 Outsourced x Log Minimum Wage; Clustered SE in Parentheses, State-Clustered SE in Brackets

	Outsourced x log(MW)	Joint Test on Leads	Trend per Decade, Without log(MW)	Trend per Decade, With log(MW)
Janitors	0.018 (0.036) [0.050]	chi2 = 1.50, p = 0.472	+0.017*** (0.006)	+0.013 (0.013)
Guards	0.009 (0.058) [0.050]	chi2 = 0.29, p = 0.864	+0.022*** (0.008)	+0.018 (0.019)

Data: CPS-ORG via IPUMS for the analysis sample; minimum-wage source: data/minimumwage_annual.csv (FRED). From '05_minwage.ipynb's interaction model (outsourced x log_mw, FE STATEFIP^YEAR + outsourced^STATEFIP + outsourced^YEAR) and trend model (outsourced x year, centered 2001, same FE minus outsourced^YEAR), both with and without the log_mw interaction. Leads: joint Wald test (chi2, df 2) that the log_mw interactions at t+1 and t+2 are zero. Trend per decade = 10 x the outsourced x year coefficient. Clustering: survey month (ym); state-clustered SE in brackets for the interaction only. Stars from the ym-clustered p-value: * p<.10, ** p<.05, *** p<.01, two-tailed. Descriptive associations, not causal.

Table A1: Pooled Wage Regression Coefficients, Janitors
Dependent Variable: Log Hourly Wage. Coefficients With SE in Parentheses

	D&K, 1983-2000	Author's Replication	Author's Extension
	Coef	1983-2000	2001-2025
Outsourced	-0.045 *** (0.005)	-0.054 *** (0.004)	-0.035 *** (0.004)
Union	0.298 *** (0.007)	0.256 *** (0.005)	0.140 *** (0.005)
Part-time	-0.173 *** (0.005)	-0.130 *** (0.005)	-0.147 *** (0.004)
Education: no schooling (ref.: college)		-0.189 *** (0.023)	-0.300 *** (0.022)
Education: primary (ref.: college)		-0.117 *** (0.011)	-0.237 *** (0.011)
Education: some high school (ref.: college)		-0.050 *** (0.011)	-0.197 *** (0.011)
Education: high school completion (ref.: college)		0.000 (0.011)	-0.108 *** (0.010)
Education: some college (ref.: college)		0.028 ** (0.011)	-0.072 *** (0.011)
Female		-0.081 *** (0.004)	-0.096 *** (0.004)
Black		-0.099 *** (0.005)	-0.096 *** (0.005)
Hispanic		-0.073 *** (0.006)	-0.057 *** (0.005)
Age		0.027 *** (0.001)	0.021 *** (0.001)
Age squared		-0.000 *** (0.000)	-0.000 *** (0.000)
MSA		0.072 *** (0.005)	0.049 *** (0.005)
Central city		-0.038 *** (0.005)	-0.030 *** (0.005)
Observations (N)	33,222	57,297	60,356
R-squared	0.44	0.40	0.40
Within R-squared		0.26	0.19

Data: CPS-ORG via IPUMS for the author's columns. D&K, 1983-2000: Dube and Kaplan's (2010) row-1 published coefficients, as published (not CPS-ORG); D&K report only outsourced, union, and parttime, so the other term rows and the R-squared are blank for that column (not reported in the paper). The author's R-squared and within R-squared come from a one-off refit of the four pooled base models (janitors/guards x 1983-2000/2001-2025), confirmed to match estimates.csv's outsourced coefficient, SE, and N exactly; D&K do not report within R-squared. Author's Replication: 1983-2000, the author's wage regression. Author's Extension: 2001-2025, estimated on its own data. The vertical rule separates the Author's Replication (1983-2000) from the Author's Extension (2001-2025). Weights: EARNWT. FE: state x year (STATEFIP^YEAR), not shown as rows. Clustering: survey month (ym), CRV1 (D&K's clustering is as published). Reference education: college. Part-time = usual hours under 30. Blank cells: term not in that model or not reported. The MSA central-city-status-unknown and geography-unidentified indicators are in the model but not shown. Stars: * p<.10, ** p<.05, *** p<.01, two-tailed; D&K stars computed here from the published SE. Table A3 gives the full D&K vs. author comparison for the outsourced coefficient.

Table A2: Pooled Wage Regression Coefficients, Guards
Dependent Variable: Log Hourly Wage. Coefficients With SE in Parentheses

	D&K, 1983-2000	Author's Replication	Author's Extension
	Coef	1983-2000	2001-2025
Outsourced	-0.202 *** (0.007)	-0.226 *** (0.007)	-0.097 *** (0.006)
Union	0.206 *** (0.011)	0.183 *** (0.010)	0.131 *** (0.010)
Part-time	-0.181 *** (0.010)	-0.224 *** (0.010)	-0.193 *** (0.009)
Education: no schooling (ref.: college)		-0.472 *** (0.081)	-0.260 (0.166)
Education: primary (ref.: college)		-0.332 *** (0.019)	-0.456 *** (0.024)
Education: some high school (ref.: college)		-0.230 *** (0.015)	-0.353 *** (0.014)
Education: high school completion (ref.: college)		-0.164 *** (0.015)	-0.219 *** (0.011)
Education: some college (ref.: college)		-0.119 *** (0.015)	-0.179 *** (0.011)
Female		-0.052 *** (0.009)	-0.083 *** (0.007)
Black		-0.059 *** (0.010)	-0.069 *** (0.008)
Hispanic		-0.084 *** (0.013)	-0.037 *** (0.009)
Age		0.038 *** (0.001)	0.031 *** (0.001)
Age squared		-0.000 *** (0.000)	-0.000 *** (0.000)
MSA		0.089 *** (0.011)	0.061 *** (0.012)
Central city		-0.032 *** (0.008)	-0.038 *** (0.007)
Observations (N)	11,116	18,251	24,143
R-squared	0.45	0.42	0.39
Within R-squared		0.28	0.20

Data: CPS-ORG via IPUMS for the author's columns. D&K, 1983-2000: Dube and Kaplan's (2010) row-1 published coefficients, as published (not CPS-ORG); D&K report only outsourced, union, and parttime, so the other term rows and the R-squared are blank for that column (not reported in the paper). The author's R-squared and within R-squared come from a one-off refit of the four pooled base models (janitors/guards x 1983-2000/2001-2025), confirmed to match estimates.csv's outsourced coefficient, SE, and N exactly; D&K do not report within R-squared. Author's Replication: 1983-2000, the author's wage regression. Author's Extension: 2001-2025, estimated on its own data. The vertical rule separates the Author's Replication (1983-2000) from the Author's Extension (2001-2025). Weights: EARNWT. FE: state x year (STATEFIP^YEAR), not shown as rows. Clustering: survey month (ym), CRV1 (D&K's clustering is as published). Reference education: college. Part-time = usual hours under 30. Blank cells: term not in that model or not reported. The MSA central-city-status-unknown and geography-unidentified indicators are in the model but not shown. Stars: * p<.10, ** p<.05, *** p<.01, two-tailed; D&K stars computed here from the published SE. Table A3 gives the full D&K vs. author comparison for the outsourced coefficient.

Table A3: D&K Pooled 1983-2000 Wage Penalty, Replication
 Outsourced Coefficient on Log Wage, Row 1 (Author's Replication vs. Reported D&K)

	Author's Replication			D&K (2010), Table 3		
	Coef		N	Coef		N
Janitors	-0.054 (0.004)	***	57,297	-0.045 (0.005)	***	33,222
Guards	-0.226 (0.007)	***	18,251	-0.202 (0.007)	***	11,116

Data: CPS-ORG via IPUMS for Author's Replication (1983-2000). D&K values are the reported row-1 estimates, as published (not CPS-ORG). Weights: EARNWT (replication choice; D&K do not state weighting). FE: state x year. Clustering: survey month (ym). Stars: * p<.10, ** p<.05, *** p<.01, two-tailed; D&K stars computed from the published SE. Author's Replication: the author's wage regression (row-1 model, no gov_employee) on 1983-2000. D&K = Dube and Kaplan's (2010) reported values.

Table A4: Robustness of the Outsourcing Wage Penalty
Outsourced Coefficient on Log Wage; SE in Parentheses; Pooled Samples

	Janitors			Guards		
	Coef		N	Coef		N
2001-2025						
Main estimate (part-time under 30 hours)	-0.035 (0.004)	***	60,356	-0.097 (0.006)	***	24,143
Part-time under 35 hours	-0.031 (0.004)	***	60,356	-0.093 (0.006)	***	24,143
Including gov_employee	-0.031 (0.004)	***	60,356	-0.067 (0.006)	***	24,143
Untrimmed wages	-0.035 (0.005)	***	60,798	-0.094 (0.006)	***	24,278
Unweighted	-0.039 (0.004)	***	60,356	-0.106 (0.005)	***	24,143
Complete cases on geography	-0.034 (0.005)	***	49,252	-0.095 (0.006)	***	20,513
Hourly-paid workers only	-0.038 (0.004)	***	50,380	-0.069 (0.006)	***	19,595
Excluding local government	-0.031 (0.004)	***	50,508	-0.097 (0.006)	***	22,754
1983-2000						
Main estimate (part-time under 30 hours)	-0.054 (0.004)	***	57,297	-0.226 (0.007)	***	18,251
Part-time under 35 hours	-0.047 (0.004)	***	57,297	-0.222 (0.007)	***	18,251
Including gov_employee	-0.053 (0.004)	***	57,297	-0.195 (0.007)	***	18,251
Untrimmed wages	-0.044 (0.005)	***	57,796	-0.222 (0.007)	***	18,286
Unweighted	-0.059 (0.004)	***	57,297	-0.227 (0.006)	***	18,251
Complete cases on geography	-0.051 (0.005)	***	47,544	-0.226 (0.007)	***	15,368
Hourly-paid workers only	-0.050 (0.004)	***	46,638	-0.202 (0.006)	***	15,058
Excluding local government	-0.052 (0.004)	***	46,062	-0.223 (0.007)	***	17,163

Data: CPS-ORG via IPUMS. Each row is one variant of the author's wage regression (base model). Panels: 2001-2025 (Author's Extension) and 1983-2000 (Author's Replication). Part-time = usual hours under 30 in the main estimate; the 35-hour variant uses the BLS threshold. Unweighted: no EARNWT weights. Untrimmed: log wage before the trimming rule. Complete cases on geography drop METRO 0, 4, 9 and the two geography indicators. Hourly-paid: PAIDHOUR == 2. Excluding local government: CLASSWKR 28 removed. N is the regression sample after pyfixest drops singleton fixed effects. Stars: * p<.10, ** p<.05, *** p<.01, two-tailed.

Table 2: Table A5: Union Premium by Outsourcing Status, by 5-Year Bin

Period	Outsourced	Union	Union $\times$ Out- sourced	N
2001-2025				
<i>Janitors</i>				
2001-2005	-0.056 (0.011)	0.169 (0.011)	-0.022 (0.026)	14,465
2006-2010	-0.062 (0.011)	0.151 (0.011)	-0.011 (0.028)	13,490
2011-2015	-0.070 (0.012)	0.150 (0.013)	0.049 (0.031)	13,422
2016-2019 [†]	-0.037 (0.009)	0.120 (0.013)	-0.033 (0.030)	9,995
2021-2025	-0.036 (0.014)	0.104 (0.014)	-0.090 (0.030)	8,984
<i>Guards</i>				
2001-2005	-0.117 (0.011)	0.117 (0.022)	0.071 (0.036)	5,426
2006-2010	-0.119 (0.014)	0.146 (0.025)	0.020 (0.035)	5,532
2011-2015	-0.147 (0.015)	0.161 (0.024)	-0.007 (0.043)	5,416
2016-2019 [†]	-0.091 (0.018)	0.158 (0.026)	-0.043 (0.035)	4,073
2021-2025	-0.085 (0.017)	0.028 (0.032)	0.059 (0.039)	3,696
1983-2000				
<i>Janitors</i>				
1983-1987	-0.079 (0.011)	0.265 (0.008)	0.024 (0.022)	17,156
1988-1992	-0.079 (0.010)	0.280 (0.010)	-0.038 (0.024)	17,300
1993-1997	-0.071 (0.010)	0.239 (0.008)	0.007 (0.024)	14,602
1998-2000 [†]	-0.062 (0.011)	0.209 (0.011)	0.000 (0.040)	8,239
<i>Guards</i>				
1983-1987	-0.308 (0.013)	0.218 (0.015)	-0.039 (0.027)	5,152
1988-1992	-0.246 (0.014)	0.240 (0.020)	-0.102 (0.029)	5,463
1993-1997	-0.229 (0.017)	0.171 (0.024)	-0.037 (0.042)	4,873
1998-2000 [†]	-0.189 (0.019)	0.104 (0.028)	-0.020 (0.048)	2,763

Interactions model (D&K row 4): the author's base model plus union_x_outsourced and parttime_x_outsourced (parttime_x_outsourced not shown). A negative Union $\times$ Outsourced coefficient means an estimated union wage premium that is smaller for outsourced workers in that bin, not a causal claim. Janitors 2021-2025 (-0.090 , SE 0.030) and guards 1988-1992 (-0.102 , SE 0.029) are significant at $p < .01$; guards 2001-2005 is marginal. Signs vary across bins, with no consistent pattern. SEs in parentheses. [†] 1998-2000 and 2016-2019 are shorter bins (3 and 4 years); 2020 is excluded from every bin. Data: CPS-ORG via IPUMS.

Table 3: Table A6: Union Mediation of the Outsourcing Wage Penalty, 1983-2000

Period	Union Rate In-House	Union Rate Out- sourced	Gap	Union Premium (β)	Raw Penalty	Union- Gap Compo- nent (log points)	% Explained	N
Janitors								
1983-1987	26.4%	18.5%	7.9pp	0.268 (0.008)	-0.074 (0.008)	-0.021 (0.003)	28.6%	17,156
1988-1992	24.0%	15.4%	8.6pp	0.276 (0.009)	-0.076 (0.009)	-0.024 (0.003)	31.0%	17,300
1993-1997	22.2%	12.9%	9.2pp	0.241 (0.009)	-0.071 (0.008)	-0.022 (0.002)	31.2%	14,602
1998-2000 [†]	20.8%	12.0%	8.7pp	0.212 (0.010)	-0.045 (0.011)	-0.018 (0.003)	40.7%	8,239
Guards								
1983-1987	23.6%	11.6%	12.0pp	0.217 (0.014)	-0.300 (0.012)	-0.026 (0.003)	8.7%	5,152
1988-1992	20.2%	10.5%	9.7pp	0.214 (0.018)	-0.249 (0.010)	-0.021 (0.003)	8.4%	5,463
1993-1997	19.2%	9.3%	9.8pp	0.162 (0.019)	-0.230 (0.013)	-0.016 (0.003)	7.0%	4,873
1998-2000 [†]	17.4%	9.4%	8.0pp	0.100 (0.024)	-0.184 (0.016)	-0.008 (0.002)	4.3%	2,763

Gap = union rate, in-house minus outsourced, in percentage points. Union premium (β) is the union coefficient from the author's base model. Raw penalty is the outsourced coefficient from the nonunion model (no union control). Union-gap component (log points) = $-(\text{gap} \times \text{union premium})$, with an approximate delta-method SE: $\sqrt{\beta^2 \text{var}(\text{gap}) + \text{gap}^2 \text{var}(\beta)}$; shown as negative, in the same direction as the raw penalty (same convention as Table 4). % explained = $|\text{union-gap component}| / |\text{raw penalty}|$, and is unstable where the raw penalty is small. SEs in parentheses. [†] 1998-2000 is a shorter bin (3 years). Descriptive decomposition, not causal. Data: CPS-ORG via IPUMS.